📘 Chapter 35- IP Phones, Voice VLANs & QoS 📡📞

🧠 1. Introduction to IP Phones (VoIP Basics)

☎️ Traditional Telephony

- Traditional phones use:
  - **PSTN (Public Switched Telephone Network)**
  - Also called **POTS (Plain Old Telephone System)**

👉 This is the old system where voice travels over **dedicated telephone lines**

🌐 IP Phones (Modern Communication)

- IP Phones use **VoIP (Voice Over IP)**
- Voice travels over an **IP network (Internet or LAN)**

👉 Means:

- Calls are sent as **data packets** 📦 over network

🔌 How IP Phones Connect

- IP Phones connect to switches like normal devices
- BUT they are **special devices** 👇

🔄 Internal 3-Port Switch inside IP Phone

An IP Phone has **3 ports**:

1. 🔼 **Uplink Port** → Connects to Switch
2. 🔽 **Downlink Port** → Connects to PC
3. 📞 **Internal Port** → Connects to Phone itself

👉 This allows:

- **PC + Phone = Single switch port usage**

💡 **Real-Life Example**

Imagine a **desk setup**:

- One cable from wall → Phone
- PC connects to phone
  👉 Saves ports + clean wiring

🔀 2. Voice VLAN Concept

🚨 **Problem**

- PC traffic + Voice traffic both go through same cable

👉 But:

- Voice needs **high priority**
- Data traffic can tolerate delay

✅ **Solution: Voice VLAN**

- Separate traffic into **2 VLANs**:
  - 🖥️ Data VLAN → PC traffic
  - 📞 Voice VLAN → IP Phone traffic

📦 **Traffic Behavior**

- PC Traffic → **UNTAGGED**
- Phone Traffic → **TAGGED with VLAN ID**

❓ **Exam Point**

👉 Why not use trunk mode?

✔ Answer:

- Because:
    - **Access VLAN + Voice VLAN = Special combination**
    - Works in **access mode only**
    - Not a normal trunk

⚙️ Configuration

interface g0/0
switchport mode access
switchport access vlan 10
switchport voice vlan 11

🔍 **Verification Commands**

show interfaces g0/0 switchport
show interfaces g0/0 trunk

💡 **Real-Life Example**

- Voice VLAN = **VIP lane on highway** 🚗
- Data VLAN = **Normal traffic lane**
  👉 VIP (voice) always gets priority

⚡ 3. Power Over Ethernet (PoE)

🔌 What is PoE?

- PoE allows:
  - **Power + Data over same Ethernet cable**

🧩 Components

- **PSE (Power Sourcing Equipment)** → Switch
- **PD (Powered Device)** → IP Phone, Camera, AP

⚙️ Working Process

1. Switch sends **low power signal**
2. Detects device requirement
3. Supplies required power only

⚠️ Why Important?

- Prevents:
  - 🔥 Device damage
  - ⚡ Overcurrent issues

💡 Real-Life Example

Like:

- Charging phone with **smart charger**
  👉 Gives only required voltage

⚠️ 4. PoE Risks & Protection (Power Policing)

🚨 Problem

- Device may draw **more power than required**
- Can damage:
  - Device 🔥
  - Switch port ⚠️

🛡️ Solution: Power Policing

⚙️ Default Behavior

power inline police

👉 Same as:

power inline police action err-disable

✓ If over power:

- Port goes **err-disabled**
- Syslog message generated

🔄 Recovery

shutdown
no shutdown

🔍 Verification

show power inline police g0/0

📢 **Alternative Mode**

power inline police action log

✓ Behavior:

- Does NOT shutdown port
- Logs message
- Restarts interface

🔢 5. PoE Standards Table

Name	Standard	Power	Wire Pairs
Cisco ILP	Non-standard	7W	2
PoE	802.3af	15W	2
PoE+	802.3at	30W	2
UPOE	802.3bt	60W	4
UPOE+	802.3bt	100W	4

📊 6. Introduction to QoS (Quality of Service)

🧠 **Old Network Design**

- Voice → PSTN
- Data → IP Network

👉 No competition → No QoS needed

🌐 **Modern Networks (Converged Networks)**

- Voice 📞
- Video 🎥
- Data 💻

👉 All share same network

🚨 **Problem**

- All traffic competes for bandwidth

✅ **Solution: QoS**

👉 QoS = Tools to:

- Control traffic
- Prioritize important packets

📊 7. QoS Key Parameters

📶 **1. Bandwidth**

- Capacity of link (bps)

👉 QoS can:

- Reserve bandwidth for voice

⏱️ 2. Delay

- Time to reach destination

Types:

- One-way delay
- Two-way delay

🔁 3. Jitter

- Variation in delay between packets

👉 IP phones use:

- **Jitter buffer** to smooth audio

📉 4. Packet Loss

- % of packets not delivered

Causes:

- Bad cables ❌
- Network congestion ❌

📏 8. QoS Standards (VERY IMPORTANT ⚠️)

Metric	Acceptable Value
One-way Delay	≤ 150 ms
Jitter	≤ 30 ms
Packet Loss	≤ 1%

🚨 **If exceeded:**

- Voice becomes:
  - Broken 🔈
  - Delayed ⏳
  - Robotic 🤖

💡 **Real-Life Example**

- Video call lagging 📞
  👉 That is:
- High delay
- High jitter
- Packet loss

🧪 9. Key Commands Summary

🔧 **Voice VLAN**

```
switchport mode access
switchport access vlan 10
switchport voice vlan 11
```

⚡ **PoE**

```
power inline police
power inline police action log
show power inline police
```

📊 QoS (Conceptual in this lecture)

- No heavy config yet
- Focus on:
  - Delay
  - Jitter
  - Loss
  - Bandwidth

⚠️ 10. Important Exam Points

✓ IP Phone has **3-port switch**
✓ Voice VLAN works with **access mode**
✓ PC traffic = **untagged**
✓ Voice traffic = **tagged**
✓ PoE supplies **power + data**
✓ QoS controls **traffic priority**

🧩 Final Understanding (Simple Way)

👉 Think of network like a **road system**:

- 🚗 Data traffic → Normal vehicles
- 🚑 Voice traffic → Ambulance

👉 QoS ensures:

- Ambulance always gets priority 🚨

🧠 ✅ 1️⃣ SUMMARY

☎️ IP Phones & VoIP

- Old system → **PSTN / POTS** 📞
- New system → **VoIP (Voice over IP)** 🌐
- Voice is sent as **data packets over network**

👉 IP Phone has **3-port switch**:

- 🔼 Uplink → Switch
- 🔽 Downlink → PC
- 📞 Internal → Phone

✓ Saves switch ports + cables

🔀 Voice VLAN

- Used to **separate Voice & Data traffic**

Traffic Type	Behavior
💻 Data (PC)	Untagged
📞 Voice	Tagged

✓ Voice VLAN gives **priority to voice traffic**

⚡ PoE (Power over Ethernet)

- Sends **Power + Data on same cable**

Device Type	Role
🔌 PSE	Switch
📱 PD	IP Phone

✓ Smart power delivery (no damage)

🛡️ Power Policing

- Prevents **over-power damage**
- Default action:
  👉 **Err-disable port**

📊 QoS (Quality of Service)

- Controls **network traffic priority**

Key Parameters:

- 📶 Bandwidth
- ⏱️ Delay
- 🔁 Jitter
- 📉 Packet Loss

Important Values:

- Delay ≤ **150 ms**
- Jitter ≤ **30 ms**
- Loss ≤ **1%**

🎯 ✅ 2️⃣ CONCLUSION

👉 Modern networks are **converged networks** (Voice + Video + Data)

✓ IP Phones + Voice VLAN ensure:

- Proper traffic separation
- Better call quality

✓ PoE simplifies infrastructure:

- No extra power cables

✓ QoS ensures:

- Voice traffic gets **highest priority**

💡 Final Understanding:
👉 Without QoS → Voice becomes **laggy, broken, robotic**
👉 With QoS → Smooth communication 🚀

🎯 Q & A

❓ Q1. What is VoIP?

✅ **Answer:**

VoIP (Voice over Internet Protocol) is a technology that allows voice communication over IP networks instead of traditional telephone lines.

👉 Voice is converted into digital packets and transmitted over network.

✓ Advantage:

- Cost-effective
- Flexible
- Scalable

❓ Q2. Difference between PSTN and VoIP?

Feature	PSTN	VoIP
Network	Telephone line	IP network
Cost	High	Low
Flexibility	Low	High

❓ **Q3. How does an IP Phone work?**

✅ **Answer:**

An IP phone:

- Converts voice → digital packets
- Sends over network
- Uses protocols like SIP

👉 It connects to switch like a normal device.

❓ **Q4. What are the ports in IP Phone?**

✅ **Answer:**

IP Phone has 3 ports:

- Uplink → Switch
- Downlink → PC
- Internal → Phone

👉 This allows PC + Phone on one cable.

❓ **Q5. What is Voice VLAN?**

✅ **Answer:**

Voice VLAN is a VLAN specifically used to carry voice traffic.

✓ It separates voice and data traffic
✓ Improves performance and quality

❓ **Q6. Why Voice VLAN is needed?**

✅ **Answer:**

Because:

- Voice requires **low delay & jitter**
- Data traffic can tolerate delay

👉 So voice traffic is prioritized.

❓ **Q7. Difference between Data VLAN and Voice VLAN?**

Feature	Data VLAN	Voice VLAN
Traffic	PC data	Voice
Tagging	Untagged	Tagged
Priority	Normal	High

❓ **Q8. Why voice VLAN is not trunk mode?**

✅ **Answer:**

Because:

- It uses **special access mode**
- Allows:
    - Untagged (PC)
    - Tagged (Phone)

👉 Not a normal trunk link.

❓ Q9. Voice VLAN configuration?

✅ **Answer:**

```
interface g0/0
switchport mode access
switchport access vlan 10
switchport voice vlan 11
```

❓ Q10. What is PoE?

✅ **Answer:**

PoE allows devices to receive **power + data over Ethernet cable**.

❓ Q11. What are PSE and PD?

✅ **Answer:**

- PSE → Power Source (Switch)
- PD → Powered Device (IP Phone, Camera)

❓ Q12. How PoE works?

✅ **Answer:**

1. Switch sends low power
2. Detects device
3. Supplies required power

👉 Prevents damage

❓ **Q13. What is Power Policing?**

✅ **Answer:**

Power policing protects devices from over-power.

👉 If device consumes more power:

- Port goes **err-disabled**

❓ **Q14. Difference between "police" and "log"?**

Mode	**Action**
police	Shutdown port
log	Only log

❓ **Q15. PoE Standards?**

Standard	**Power**
802.3af	15W
802.3at	30W
802.3bt	60–100W

❓ **Q16. What is QoS?**

✅ **Answer:**

QoS (Quality of Service) is a technique used to prioritize important traffic like voice and video.

❓ Q17. Why QoS is required?

✅ **Answer:**

Because:

- Multiple traffic types share same network
- Voice needs priority

❓ Q18. What are QoS parameters?

✅ **Answer:**

- Bandwidth
- Delay
- Jitter
- Packet Loss

❓ Q19. What is Jitter?

✅ **Answer:**

Jitter is variation in packet delay.

👉 High jitter = bad voice quality

❓ Q20. What is acceptable delay?

✅ **Answer:**

- ≤ 150 ms

❓ **Q21. What happens if QoS not applied?**

✅ **Answer:**

- Voice becomes:
  - Broken 🔊
  - Delayed ⏳
  - Robotic 🤖

🧪 🔹 **PRACTICAL QUESTIONS**

❓ **Q22. How to verify Voice VLAN?**

✅ **Answer:**

show interfaces g0/0 switchport

❓ **Q23. How to check PoE status?**

✅ **Answer:**

show power inline

❓ **Q24. How to recover err-disabled port?**

✅ **Answer:**

shutdown
no shutdown

❓ Q25. Troubleshooting: Phone not working?

✅ **Answer:**

Check:
✓ VLAN config
✓ PoE power
✓ Cable
✓ QoS

❓ Q26. Why voice is breaking during calls?

✅ **Answer:**

Possible reasons:

- High delay
- High jitter
- Packet loss
- No QoS

❓ Q27. Explain full packet flow?

✅ **Answer:**

1. Phone generates voice
2. Tagged with VLAN
3. Switch forwards
4. QoS prioritizes
5. Destination receives

🎯 🧠 One Liner

👉 “Explain complete Voice VLAN + QoS setup”

Best Answer:

- Configure access + voice VLAN
- Ensure PoE enabled
- Apply QoS for priority
- Verify with commands

💡 Say this confidently → You’ll look **PRO level** 🔥

🚀 FINAL TAKEAWAY

✓ IP Phone = Smart network device
✓ Voice VLAN = Traffic separation
✓ PoE = Power + Data
✓ QoS = Priority control

👉 Together they ensure:
🎯 **Clear, fast, and reliable communication**
